

FINAL PROJECT PRESENTATION

Car Diagnosis Expert System using Prolog

Artificial Intelligence CCE414 | May 2026

Ramy Said
231903795

David George
231903654

Ahmed Mohamed
232903959



The Introduction

Problem Statement and Project Vision

Problem Statement

The Core Challenge

Car owners often face sudden vehicle malfunctions but lack the mechanical expertise to identify root causes.

This leads to:



Unnecessary stress and safety hazards.



Reliance on costly diagnostic services for simple fixes.



Long waiting times for professional inspections.



System Purpose & Target Users



Core Purpose

Design an **expert system** to diagnose common problems via **yes/no questions**.

It identifies faults and suggests repairs efficiently for various vehicle issues.



Primary Users

Focused on **everyday car owners**, drivers, and **novice mechanics**.

Designed for individuals requiring immediate troubleshooting assistance without advanced technical knowledge.

Resources & Expertise



Software

Developed using **SWI-Prolog** for the robust knowledge base.



Knowledge

Modeled after **automotive manuals** and diagnostic trees.



Emulation

Mimics the reasoning of an **experienced auto-mechanic**.



Knowledge Engineering

Rule-Based Diagnosis Logic

Diagnostic Knowledge Base (1-5)



1. Battery

If engine **won't start** and lights are **dim**.



2. Spark Plugs

If engine **misfires** and **high fuel consumption**.



3. Radiator

If car is **overheating** and **leaking coolant**.



4. Brake Pads

If **squealing brakes** and **poor braking performance**.



5. Flat Tire

If **flapping noise** and car **pulls to side**.

Diagnostic Knowledge Base (6–10)



6. Fuel Pump

If engine **stalls** and **whining noise** exists.



7. Oxygen Sensor

If **check engine light** on and **black smoke**.



8. Alternator

If **battery light on** and **electrical issues**.



9. Transmission

If **hard shifting** and **leaking red fluid**.



10. Worn Wiper

If **poor visibility** and **streaking on glass**.

System Architecture

The system is built on **Logic Programming** using **Prolog**:



Backward Chaining

Logic travels from Goal to Facts.



The Cut Operator (!)

A crucial mechanism that prevents backtracking once a match is found.



Recursion

Used to traverse the Knowledge Base efficiently.

```
PROJECT.pl
File Edit Browse Compile Prolog Help
PROJECT.pl

start :-
    write('*** Car Diagnosis System ***'), nl,
    write('Answer the questions with y. or n. '), nl,
    write('-----'), nl,
    diagnose(Problem), !,
    write('>> Diagnosis: The problem is '), write(Problem), nl,
    write('>> Advice: '), solution(Problem), nl.

start :-
    write('Sorry, the system cannot identify the problem.'), nl.

diagnose(battery) :- ask(engine_wont_start), ask(dim_lights).
diagnose(spark_plugs) :- ask(engine_misfires), ask(high_fuel_consumption).
diagnose(radiator) :- ask(overheating), ask(leaking_coolant).
diagnose(brake_pads) :- ask(squealing_brakes), ask(poor_braking).
diagnose(flat_tire) :- ask(flapping_noise), ask(car_pulls_to_side).
diagnose(fuel_pump) :- ask(engine_stalls), ask(whining_noise).
diagnose(oxygen_sensor) :- ask(check_engine_light), ask(black_smoke).
diagnose(alternator) :- ask(battery_light_on), ask(electrical_issues).
diagnose(transmission) :- ask(hard_shifting), ask(leaking_red_fluid).
diagnose(worn_wiper) :- ask(poor_visibility), ask(streaking_glass).

solution(battery) :- write('Replace the battery.').
solution(spark_plugs) :- write('Clean or replace spark plugs.').
solution(radiator) :- write('Check hoses and refill coolant.').

usersolution/1: (loaded) 10 clauses, 2.2 Kb, primary_index(1)
```

Diagnostic Algorithm

The algorithm follows a linear sequence from initialization to output, with a specific mechanism for handling rule matches.

If the rule matches, the **Cut operator** stops the search. If no rules match, the **fallback predicate** informs the user.



1. Start

Initialize **start**



2. Ask

Evaluate **Diagnose Rules**



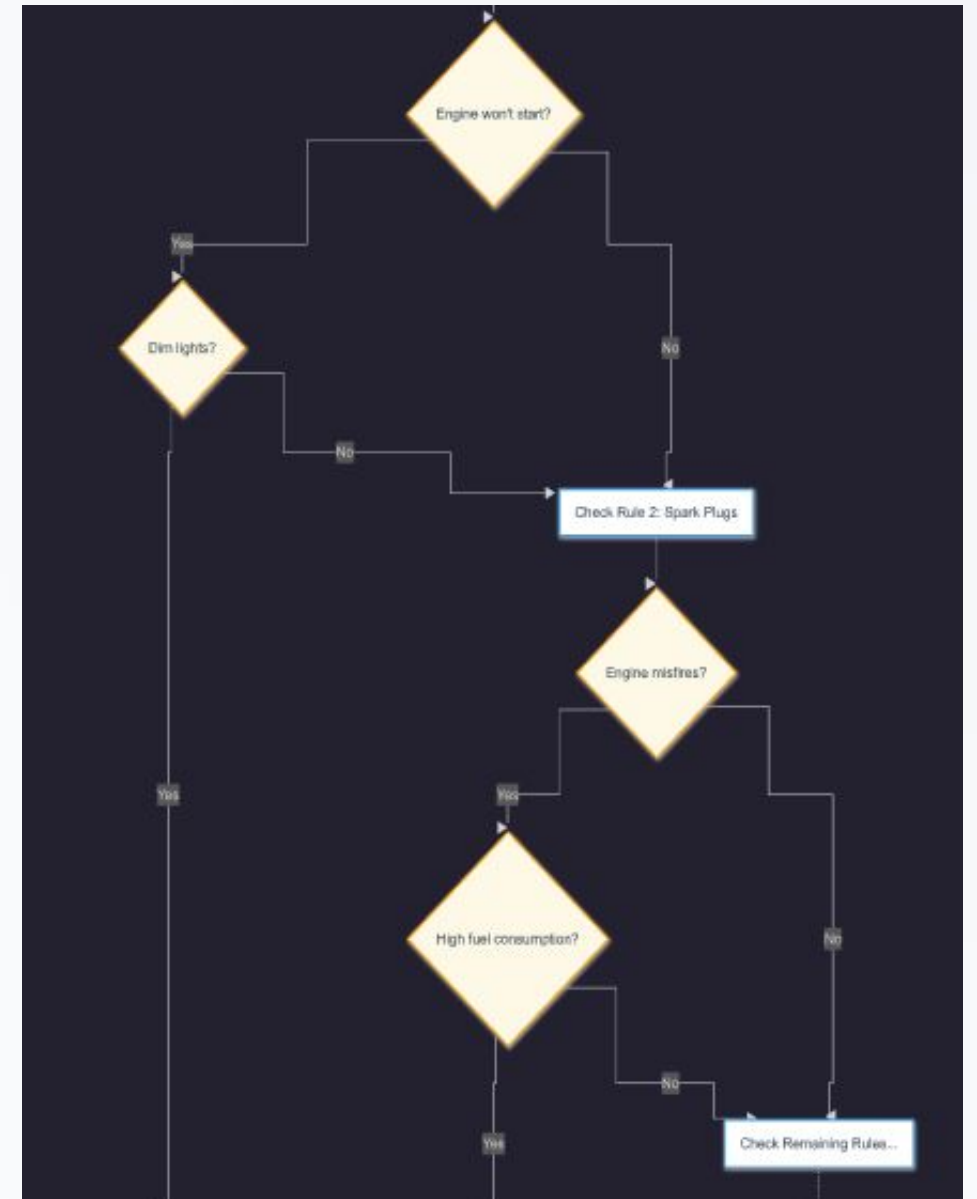
3. Process

User Input (**y/n**)



4. Output

Print Advice



System Output & Results

Diagnosis Result	User Symptoms Provided	Recommended Advice
Battery	Engine won't start, Dim lights	Replace the battery
Spark Plugs	Engine misfires, High consumption	Clean/Replace spark plugs
Radiator	Overheating, Leaking coolant	Check hoses/refill coolant
Unsuccessful	No matching symptoms	System cannot identify

System Output & Results

Diagnosis Result	User Symptoms Provided	Recommended Advice
Battery	Engine won't start; Dim lights	Replace the battery
Spark Plugs	Engine misfires; High consumption	Clean/Replace spark plugs
Radiator	Overheating; Leaking coolant	Check hoses/refill coolant
Unsuccessful	No matching symptoms	<i>System cannot identify</i>

Thank You!

Questions?

Artificial Intelligence | Engineering CCE414

5 / 10 / 2026